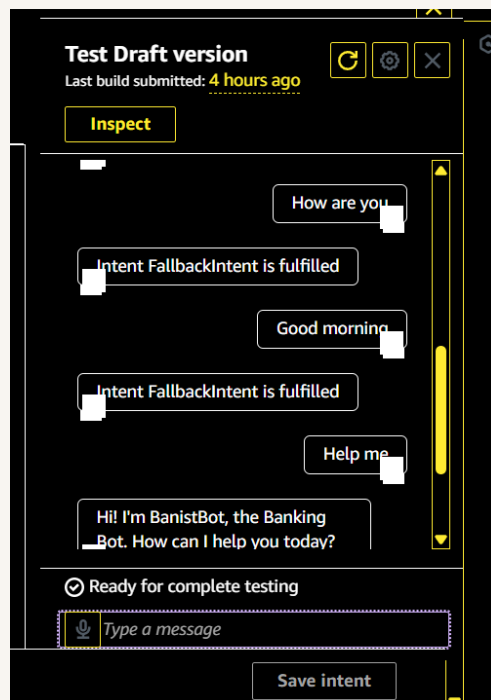




Build a Chatbot with Amazon Lex

J Michael K.





Introducing Today's Project!

Project overview

In this project, I will create a chatbot called DanBot, designed for an imaginary banking service. This bot will greet users and be ready to handle conversations through voice or text.

What is Amazon Lex?

Amazon Lex is a fully managed artificial intelligence (AI) service from AWS that lets you build voice and text conversational interfaces using Natural language.

Key tools and concepts

Services I used were Amazon Lex and IAM. Key concepts I learnt include : Intent, Utterances, Intent variations, Fallbackintent and Intent classification confidence score

Personal reflections

One thing I didn't expect in this project was for the steps by steps implementation to be that detailed

Project timeline

This project took me approximately 45 Mins



Setting up a Lex chatbot

Approach to chatbot setup

In this step, I will create a new bot from scratch on Amazon Lex.

Chatbot creation process

I created my chatbot from scratch with Amazon Lex. Setting it up took me 10 mins

IAM role configuration

While creating my chatbot, I also created a role with basic permissions because i want to allow Amazon Lex to call other AWS services on my behalf and since i will need it to interact with AWS Lambda later i had to do it.

Intent classification confidence score

In terms of the intent classification confidence score, I kept the default value of 0.40. This means that my chatbot needs to be at least 40% confident that it understands what the user is asking to be able to give a response. So if a user's input is ambiguous and my chatbot's confidence score is below 0.4, it'll throw an error message.



▼ Language: English (US)

Select language

English (US) ▼

Description - optional

Interactive Language

Maximum 2000 characters.

☐ Enable speech-to-speech inference

Voice interaction

The text-to-speech voice that your bot uses to interact with users.

Gregory ▼

Voice sample

Hello, my name is Gregory. Let me know how I can assist you.

Play

Speech model preference

The preferred speech model to use for speech-to-text

None ▼

Intent classification confidence score threshold

0.40

Min: 0.00, max: 1.00.



Intents

In this step, I will train BanisBot to greet users.

Creating my first intent

Intents are what the user is trying to achieve in their conversation with the chatbot. For example, checking a bank account balance; booking a flight; ordering food.

I created my first intent, WelcomeIntent, to allow the BanisBot to greet the users.

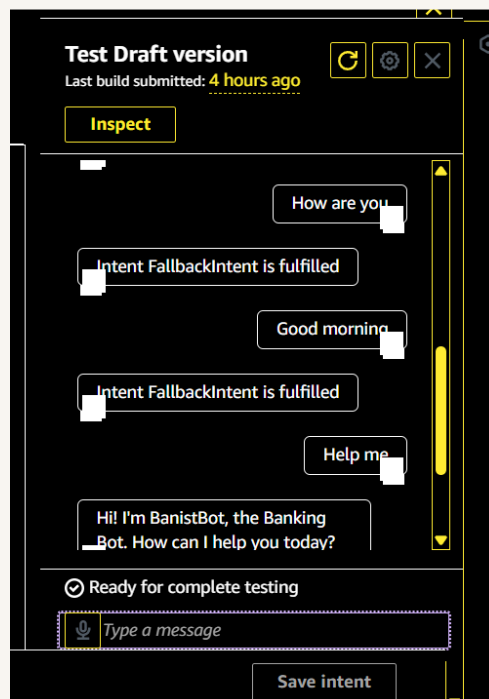
Testing chatbot responses

I launched and tested my chatbot, which could respond successfully if I enter the greetings options i defined in the Utterances section.



Handling unrecognized inputs

My chatbot returned the error message 'Intent FallbackIntent is fulfilled' when I entered any other greetings kind which are not part of the defined one in the Utterances section. This error message occurred because of the inability of the bot to understand the inputs (With an intent classification confidence score of less than 0.40), thus generating the error.





Configuring FallbackIntent

In this step, I will customise FallbackIntent to send user-friendly messages.

When FallbackIntent triggers

FallbackIntent is a default intent in every chatbot that gets triggered when my chatbot has a confidence score below 40% (0.4) for all the intents I've defined (in my case, it's just the WelcomeIntent for now)

I wanted to configure FallbackIntent because i want my users to receive a dynamic range of responses, making them sound more conversational.

My FallbackIntent configuration

To configure FallbackIntent, I changed FallbackIntent's closing response to: " Sorry I am having trouble understanding. Can you describe what you'd like to do in a few words? I can help you find your account balance, transfer funds and make a payment.".



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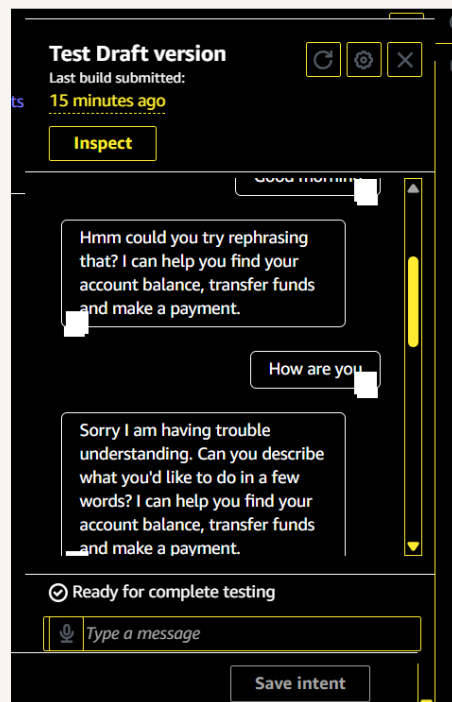
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I also added variations! What this means for an end user is that he/she will receive different responses of the same kind (same sense) from the bot, when the bot doesn't understand his/ her responses, making their interaction sound more conversational.



FallbackIntent with Variations





Initial Responses

In this project extension I'm about to spice up FallbackIntent with an initial response and add variations to the initial response.

Setting up initial responses

Initial responses are greeting or confirmation my chatbot sends right after it figures out my user's intent. Lex would send the initial response at the very start of the intent's flow, before anything else I've set up in my intent. I've set up initial responses to give to my users a heads up about what to expect.

Initial response implementation

The initial response messages I set up are : Hmmmm this is interesting..., One moment..., and let me think... For the user, this means that the bots has acknowledged their reponses / meaasage.

